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cells are elongated and possesses a large central vacuole, many nuclei and numerous discoidal chloroplasts.

The cytoplasm is divided into outer ectoplasm and inner endoplasm. The endoplasm shows cytoplasmic streaming.

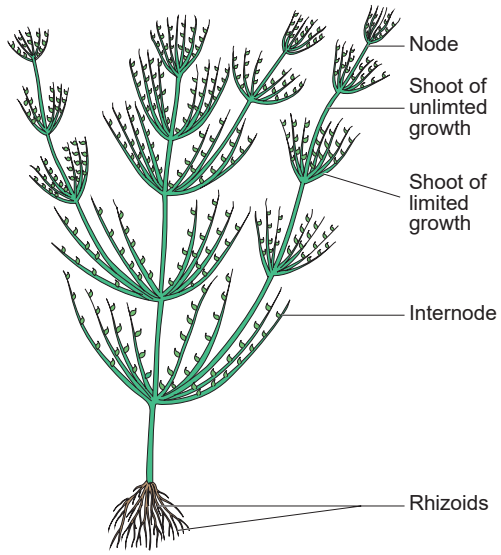


Figure 2.8: *Chara* Habit

Reproduction

Chara reproduces by vegetative and sexual methods. Vegetative reproduction takes place by Amylum stars, Root bulbils, Amorphous bulbils and secondary protonema.

Sexual reproduction - Sexual reproduction is Oogamous. Sex organs are macroscopic and are produced on the branches of limited growth. The male sex organ is called Antheridium or Globule and the female sex organ is called Oogonium or Nucule (Figure 2.9). The Nucule is located above the Globule. The antheridium is spherical, macroscopic and its wall is made up of eight cells called shield cells. The antheridium has spermatogenous filaments. These filaments produce antherozoids. The nucule is covered by five spirally twisted tube cells and five coronal cells are present at the top of the nucule (Figure 2.9). The centre of the nucule

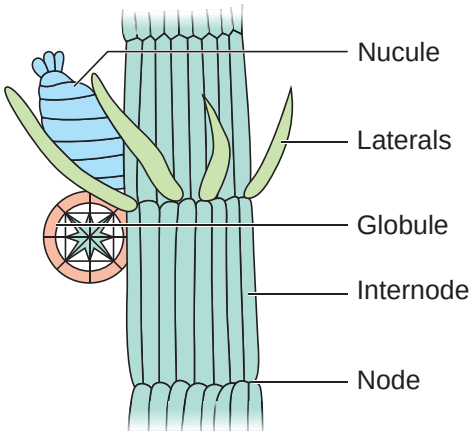


Figure 2.9: *Chara* sex organs

possesses a single egg. At maturity the tube cells separate and a narrow slit is formed. The antherozoids penetrate the oogonium and one of them fuses with the egg to form a diploid oospore. The oospore secretes a thick wall around and germinates after the resting period. The nucleus of the oospore divides to form 4 haploid daughter nuclei of which, three degenerate. The oospore or zygote germinates to produce haploid protonema. The plant body of *Chara* is

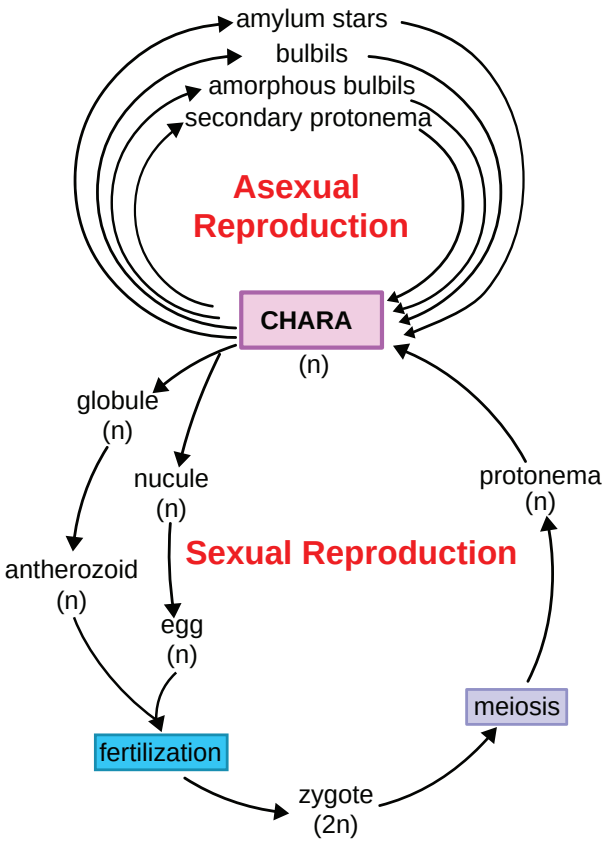


Figure 2.10: Life cycle of *Chara*